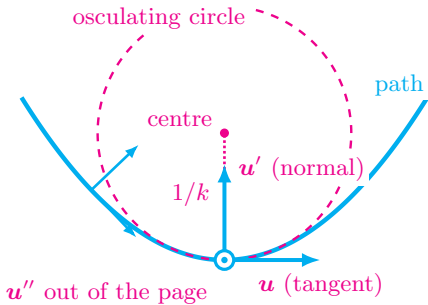


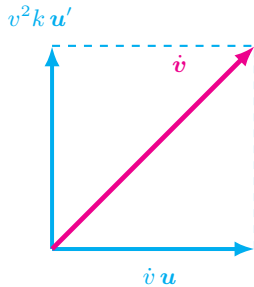
The path sets the local axes



Planar example: $\tau = 0$

Acceleration: speed and turning

changes direction



changes speed

$$\dot{\mathbf{v}} = \dot{v} \mathbf{u} + v^2 k \mathbf{u}'$$